Hackathon Proposal: StockSync AI for Orbyte

Autonomous Inventory Forecasting & Purchase Order Dispatch with Human-in-the-Loop Governance

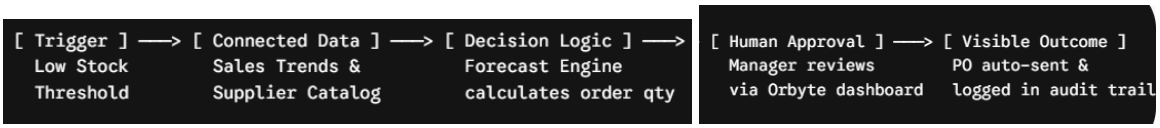
1. Problem Statement

Malaysian retail and F&B micro-SMEs frequently experience stockouts and over-ordering caused by disjointed tracking across paper notes, WhatsApp chats, and disconnected spreadsheets. Business owners spend hours every week calculating buffer inventory, chasing supplier price sheets, and manually drafting purchase orders (POs). Errors in this manual workflow directly impact working capital, cause stock shortages, and create operational stress.

2. Solution Overview

StockSync AI is a modular procurement and inventory automation plugin for the Orbyte shared business hub. It continuously tracks SKU burn rates against lead times, automatically predicts stockout windows, drafts supplier purchase orders with AI-calculated quantities, and routes them to managers for a single-click review before dispatching via WhatsApp or email.

3. Core Automated Journey (The 5 Steps)



1. **Trigger:** Daily automated background check, or an inventory level reaching a dynamic reorder safety buffer.
2. **Connected Data:** Ingests synthetic POS transaction history, current SKU quantities, supplier contact books, minimum order quantities (MOQs), and standard lead times.
3. **Decision Logic (AI):** A time-series demand forecasting model (or regression with seasonality heuristics) calculates the optimal order date and stock quantity. It selects the preferred supplier and prepares a complete Purchase Order draft.
4. **Human Approval:** The system flags an alert to the Store Manager on Orbyte: *"Stock for Item X will deplete in 4 days. Recommended reorder: 120 units from Supplier Y."* The manager can approve, adjust quantity, or reject.
5. **Visible Outcome & Audit Log:** Upon approval, a finalized PDF PO is generated and sent to the supplier via automated email/mock WhatsApp API. The transaction is recorded in Orbyte's immutable activity audit log, updating status to Pending Delivery.

4. Alignment with Rubric & Must-Have Requirements

Requirement	How StockSync AI Delivers
Clear SME Problem	Eliminates manual stock calculations and prevents revenue loss from stockouts.

People Stay in Control	Role-based permissions (only Store Managers can approve purchases), hard financial threshold warnings, and full audit logs.
Ease of Learning	Simple 1-click approve/edit card requiring zero technical configuration or training.
Orbyte Fit & Reusability	Acts as a plug-and-play module that ties Orbyte's core business directory, contact list, and notifications together across retail, grocery, and F&B SMEs.
Measurable Business Impact	Saves 5–8 hours/week on manual stock management, reduces stockout frequency by an estimated 35%, and eliminates ordering calculation errors.
SDG Alignment	**SDG 8 (Target 8.2 & 8.3):** Boosts SME productivity through digital automation. **SDG 12 (Target 12.3 & 12.5):** Reduces inventory spoilage and dead-stock waste.

5. Technical Implementation & Feasibility (Preliminary Round)

- **Frontend:** Clean web interface built with React / Next.js / Streamlit showing the inventory monitor, forecasting visualizer, and single-click approval inbox.
- **Backend:** Python (FastAPI) or Node.js handling the automation logic and simulated Orbyte event triggers.
- **Data & AI Layer:** Synthetic dataset (JSON/SQLite) containing 90 days of retail transactions, SKUs, and supplier records. Lightweight demand forecasting using Prophet, scikit-learn, or an LLM structured function-calling prompt to output JSON-based PO drafts.
- **Notification / Dispatch:** Mock email/WhatsApp dispatch with status callback confirming delivery.

6. Key Risk & Fallback Strategy

- **Risk:** Inaccurate AI demand forecasts due to erratic sales spikes or seasonal anomalies, risking cash flow lockup.

- **Fallback & Guardrails:** Hard safety guardrails (e.g., maximum budget caps per order, sanity checks capping orders at $2 \times$ average historical batch size) and explicit human approval required before any financial commitment is dispatched.